Diagonalization, $\vec{\S}$5.2

Recall from last class:

- If A is an $n \times n$ matrix, then $\lambda \in \mathbb{R}$ is an **eigenvalue** of A if there exists a non-zero vector v, called an **eigenvector** corresponding to λ, such that
$$Av = \lambda v.$$
- If λ is an eigenvalue of an $n \times n$ matrix A, then the set of all solutions v to $Av = \lambda v$ is the **eigenspace** of A corresponding to λ.
- **Theorem 1:** The eigenvalues of a triangular matrix are the diagonal entries.
- **Theorem 2:** If $v_1, \ldots, v_p$ are eigenvectors corresponding to distinct eigenvalues $\lambda_1, \ldots, \lambda_p$ of an $n \times n$ matrix A, then $v_1, \ldots, v_p$ are linearly independent.
- If A is an $n \times n$ matrix, then the characteristic polynomial of A, denoted by $p_A(\lambda)$, is given by
$$p_A(\lambda) = \det(A - \lambda I_n).$$
- λ is an eigenvalue of an $n \times n$ matrix A if and only if λ is a zero of the characteristic polynomial.
- If A and B are similar matrices, then they have the same characteristic polynomials and hence the same eigenvalues (with the same algebraic multiplicities).

1. Let $A = \begin{bmatrix} 1 & 1 \\ -2 & 4 \end{bmatrix}$. Find eigenvalues and eigenspaces of A.

Definition 1. *An $n \times n$ matrix A is* **diagonalizable** *if A is similar to a diagonal matrix.*

Diagonalization Theorem: An $n \times n$ matrix A is diagonalizable if and only if A has n linearly independent eigenvectors. Moreover, $A = PDP^{-1}$ with D diagonal if and only if the columns of P are n linearly independent eigenvectors of A. In this case the diagonal entires of D are the corresponding eigenvalues of A.

Definition 2. *A basis of* $\mathbb{R}^n$ *diagonalizing* A *is called an* **eigenbasis**.

2. Prove the Diagonalization Theorem.

3. Let $A = \begin{bmatrix} 1 & 3 & 3 \\ -3 & -5 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize A.

4. Let $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$ If possible, diagonalize A.

Theorem 6: An $n \times n$ matrix A with n distinct eigenvalues is diagonalizable.

5. Prove the theorem.